

Assignment 1 Solutions

Q1. Limitations of Classification Accuracy

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$.

Accuracy fails in imbalanced datasets because majority class dominates evaluation.

Example: If 990 negatives and 10 positives exist, predicting all negative gives 99% accuracy but zero useful detection.

Accuracy paradox occurs when high accuracy hides poor minority-class performance.

Alternative metrics include Precision, Recall, F1-score, Balanced Accuracy, and MCC.

Q2. Precision, Recall, F1-score, Specificity

Precision = $TP / (TP + FP)$. Recall = $TP / (TP + FN)$. Specificity = $TN / (TN + FP)$.

F1-score = $2PR / (P + R) = 2TP / (2TP + FP + FN)$.

Precision penalizes false positives; recall penalizes false negatives.

In medical diagnosis recall is often critical because missing disease is costly.

Q3. ROC Curve

ROC plots True Positive Rate against False Positive Rate.

TPR = $TP / (TP + FN)$, FPR = $FP / (FP + TN)$.

Changing threshold changes both TPR and FPR.

AUC measures probability that a positive sample ranks above a negative sample.

Q4. Parametric vs Non-Parametric Models

Parametric models assume fixed functional form, e.g., logistic regression.

Non-parametric models grow complexity with data, e.g., k-NN.

Parametric models: high bias, low variance.

Non-parametric models: low bias, high variance.

Q5. Overfitting and Underfitting

Underfitting occurs when model is too simple; overfitting when model memorizes noise.

Bias-variance decomposition: $\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$.

Cross-validation estimates generalization error and helps choose proper complexity.